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                        FILE MANIFEST & QUICK REFERENCE
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PROJECT: Multi-Drone Digital Array Radar Simulation @ 10 GHz
VERSION: 1.0 (2026)
STATUS: Ready for Educational & Research Use

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CORE SIMULATION FILES (3 MATLAB Scripts)
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1. multi_drone_radar_main.m
   ├─ PURPOSE: Main simulation engine
   ├─ SIZE: ~1,100 lines of code
   ├─ RUNTIME: 10-15 minutes (10-second simulation)
   ├─ OUTPUTS: 7 analysis plots
   │  ├─ Range tracking (4 drones + detections)
   │  ├─ Radial velocity profiles
   │  ├─ Micro-Doppler spectrogram
   │  ├─ SNR evolution
   │  ├─ 3D flight paths
   │  ├─ Beamforming steering angles (Az & El)
   │  └─ Radiation patterns (4 snapshots)
   ├─ KEY FUNCTIONS USED:
   │  ├─ multiDroneMotion() × 4 per pulse
   │  ├─ phased.LinearFMWaveform (LFM generation)
   │  ├─ phased.Transmitter & phased.Receiver
   │  ├─ phased.FreeSpace (propagation)
   │  ├─ phased.RadarTarget (RCS model)
   │  ├─ phased.PhaseShiftBeamformer (digital beamforming)
   │  ├─ phased.MatchedFilter (range compression)
   │  └─ Various signal processing functions
   ├─ CONFIGURATION OPTIONS:
   │  ├─ Ndrones (default: 4)
   │  ├─ totalDuration (default: 10 seconds)
   │  ├─ fc (default: 10e9 Hz)
   │  ├─ ura.Size (default: [8 8])
   │  └─ sweepBW, prf, fs (various radar params)
   ├─ MEMORY USAGE: ~5 GB
   └─ HOW TO RUN:
      >> multi_drone_radar_main
      % Wait for simulation to complete (shows progress every 1000 pulses)
      % 7 figures generated automatically

2. multiDroneMotion.m
   ├─ PURPOSE: Drone kinematics helper function
   ├─ SIZE: ~130 lines of code
   ├─ INPUTS:
   │  ├─ t (current simulation time, seconds)
   │  ├─ dt (time step, 1/PRF)
   │  ├─ tgtmotion (phased.Platform object)
   │  ├─ radarpos (radar position, 3×1)
   │  ├─ rotorOffset (rotor hub offsets, 3×4)
   │  ├─ rotorRadius (blade length, meters)
   │  ├─ bladeRate (rotor RPM converted to rad/s)
   │  └─ bladePhase (initial phase per rotor)
   ├─ OUTPUTS:
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   │  ├─ scatterPos (3×9 positions: body + 8 blade tips)
   │  ├─ scatterVel (3×9 velocities)
   │  └─ scatterAng (2×9 azimuth/elevation angles)
   ├─ PHYSICS MODELED:
   │  ├─ Kinematic body motion (from phased.Platform)
   │  ├─ Circular blade rotation (x-y plane)
   │  ├─ Blade tip velocity (translation + rotation)
   │  └─ Look angle computation
   ├─ CALLS PER ITERATION:
   │  - Called once per drone per pulse
   │  - 4 drones × 50,000 pulses = 200,000 calls
   └─ HOW IT'S USED:
      [scatterPos, scatterVel, scatterAng] = multiDroneMotion(
         t, dt, tgtmotion{d}, radarpos, rotorOffset,
         rotorRadius, bladeRate, bladePhase)

3. beamforming_analysis.m
   ├─ PURPOSE: Advanced beamforming characterization
   ├─ SIZE: ~700 lines of code
   ├─ RUNTIME: 2-3 minutes
   ├─ OUTPUTS: 9 comprehensive analysis figures
   │  ├─ 4×1 Individual beam patterns (one per drone)
   │  ├─ 3D radiation pattern (full surface)
   │  ├─ Multi-beam gain response (overlay)
   │  ├─ 4×1 Polar radiation patterns
   │  ├─ On-axis vs off-axis gain comparison
   │  ├─ Spatial resolution & beamwidth analysis
   │  ├─ Beam steering dynamics (heatmap)
   │  ├─ Adaptive weight visualization (magnitude & phase)
   │  └─ Multi-beam simultaneous tracking capability
   ├─ ANALYSIS PROVIDED:
   │  ├─ Gain (on-axis & off-axis, dB)
   │  ├─ Sidelobe suppression (dB)
   │  ├─ 3dB beamwidth (degrees)
   │  ├─ Weight magnitude/phase per element
   │  └─ Interference rejection metrics
   ├─ DEPENDENCIES:
   │  ├─ phased.URA (array geometry)
   │  ├─ phased.SteeringVector (phase shifts)
   │  └─ pattern() function (MATLAB built-in)
   ├─ KEY METRICS COMPUTED:
   │  ├─ Theoretical beamwidth: 1-5° depending on array size
   │  ├─ On-axis gain: ~18 dB (64-element coherent)
   │  ├─ Sidelobe level: ~-20 dB (uniform weighting)
   │  └─ Interference between drones: 20-40 dB suppression
   └─ HOW TO RUN:
      >> beamforming_analysis
      % Runs independently (creates own array configuration)
      % 9 figures generated with detailed beamforming metrics

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DOCUMENTATION FILES (5 Markdown/Text Files)
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1. README.md
   ├─ PURPOSE: Comprehensive system documentation
   ├─ SIZE: ~600 lines
   ├─ SECTIONS:
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│   ├─ 1. Overview (key features, system architecture)
│   ├─ 2. System Architecture (radar specs, drone specs, array config)
│   ├─ 3. Files Description (detailed explanation of each .m file)
│   ├─ 4. Digital Beamforming Explanation (theory & practice)
│   ├─ 5. Simulation Workflow (pulse-by-pulse processing)
│   ├─ 6. Key Outputs & Interpretation (plot analysis guide)
│   ├─ 7. Advanced Features (adaptive algorithms, micro-Doppler, etc.)
│   ├─ 8. References (books, papers, online resources)
│   └─ Appendices (glossary, FAQs)
├─ BEST FOR:
│   ├─ Understanding overall system design
│   ├─ Learning radar theory & signal processing
│   ├─ Detailed plot interpretation
│   ├─ Advanced topics (mutual coupling, phase noise, etc.)
│   └─ Research & academic reference
└─ READ TIME: 30-45 minutes (full), 10 min (key sections)

2. QUICKSTART.md
├─ PURPOSE: User-friendly installation & usage guide
├─ SIZE: ~450 lines
├─ SECTIONS:
│   ├─ Installation & Setup (5 minutes)
│   ├─ Running the Simulation (3 options)
│   ├─ Parameter Configuration (4 preset configs)
│   ├─ Understanding the Output (plot interpretation)
│   ├─ Beamforming Analysis Output (9 figures explained)
│   ├─ Key Metrics Interpretation (range res, beamwidth, etc.)
│   ├─ Troubleshooting Guide (common issues & solutions)
│   ├─ Performance Tips (speed vs accuracy trade-offs)
│   └─ MATLAB Cheat Sheet (useful commands)
├─ BEST FOR:
│   ├─ First-time users (start here!)
│   ├─ Quick parameter tuning
│   ├─ Understanding output plots
│   ├─ Troubleshooting errors
│   └─ Learning by experimentation
├─ PRESET CONFIGURATIONS PROVIDED:
│   ├─ Config A: Quick test (2 min) - small array, short duration
│   ├─ Config B: Balanced (5 min) - standard setup [DEFAULT]
│   ├─ Config C: High-resolution (15 min) - large array, long duration
│   └─ Config D: Low-frequency (5 min) - 5 GHz instead of 10 GHz
└─ READ TIME: 15-20 minutes (setup), refer as needed

3. BEAMFORMING_REFERENCE.txt
├─ PURPOSE: Technical reference for signal processing theory
├─ SIZE: ~500 lines (very detailed)
├─ SECTIONS:
│   ├─ 1. Digital Array Fundamentals (basic concepts)
│   ├─ 2. Steering Vector (phase alignment theory)
│   ├─ 3. Beamforming Operation (matched weighting)
│   ├─ 4. Beam Pattern Visualization (gain plots, beamwidth)
│   ├─ 5. Adaptive Beam Steering (per-pulse steering algorithm)
│   ├─ 6. Multi-Drone Tracking Strategy (closest-first policy)
│   ├─ 7. Beamforming Gain Analysis (SNR improvement metrics)
│   ├─ 8. Beamwidth & Spatial Resolution (angular separation)
│   ├─ 9. Weight Magnitude & Phase (visualization explanation)
│   ├─ 10. Adaptive Algorithms (MVDR, LCMV, RLS, SMB - not implemented)
│   ├─ 11. Frequency-Dependent Steering (LFM chirp effects)
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│   ├─ 12. Matched Filter Stage (range compression)
│   ├─ 13. Micro-Doppler Modulation (blade rotation signatures)
│   ├─ 14. Computational Cost (runtime & memory analysis)
│   └─ 15. Practical Considerations (real-world effects)
├─ MATHEMATICAL CONTENT: High (equations & derivations)
├─ BEST FOR:
│   ├─ Understanding beamforming mathematics
│   ├─ Steering vector derivation & interpretation
│   ├─ Micro-Doppler physics explanation
│   ├─ Algorithm details & implementation
│   ├─ Advanced signal processing concepts
│   └─ Research & technical validation
├─ EQUATIONS PROVIDED:
│   ├─ Steering vector formula
│   ├─ Beamformer output equation
│   ├─ Beam pattern response
│   ├─ Array gain formula
│   ├─ Beamwidth approximation
│   ├─ Radar equation (range dependence)
│   ├─ Doppler & micro-Doppler formulas
│   └─ Computational complexity analysis
└─ READ TIME: 45-60 minutes (complete), reference as needed

4. SYSTEM_OVERVIEW.txt
   ├─ PURPOSE: Complete project summary & file navigation guide
   ├─ SIZE: ~550 lines
   ├─ SECTIONS:
   │   ├─ Project Summary (what this is, who should use it)
   │   ├─ Deliverable Files (3-line descriptions of each file)
   │   ├─ Quick Start (5 minutes to first results)
   │   ├─ System Specifications (radar, antenna, waveform params)
   │   ├─ Drone Specifications (geometry, RCS, blade physics)
   │   ├─ Simulation Parameters (duration, pulses, memory, CPU)
   │   ├─ Output Plots (all 7 main figures explained)
   │   ├─ Beamforming Analysis (9 additional figures)
   │   ├─ Physics Models (what's implemented, what's simplified)
   │   ├─ Parameter Tuning (quick configs for different scenarios)
   │   ├─ Typical Results Interpretation (what to expect)
   │   ├─ Advanced Analysis (micro-Doppler, velocity unfolding, etc.)
   │   ├─ Troubleshooting Guide (common issues)
   │   ├─ Performance Metrics (detection range, resolution, etc.)
   │   ├─ References & Further Reading
   │   ├─ File Navigation Guide (where to look for what)
   │   ├─ Troubleshooting Guide (detailed issue resolution)
   │   └─ Summary Checklist (pre/during/after simulation)
   ├─ BEST FOR:
   │   ├─ Project overview (read first!)
   │   ├─ File navigation & quick reference
   │   ├─ Understanding what each output means
   │   ├─ Performance metric interpretation
   │   ├─ Troubleshooting & support
   │   └─ Choosing appropriate parameter configuration
   └─ READ TIME: 20-30 minutes (reference as needed)

5. FILE_MANIFEST.txt (THIS FILE)
   ├─ PURPOSE: Detailed description of all files & how to use them
   ├─ SIZE: ~300 lines
   ├─ CONTENT:
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```
    │   ├─ This file itself (meta!)
    │   ├─ Descriptions of all 3 MATLAB scripts
    │   ├─ Descriptions of all 5 documentation files
    │   ├─ Quick reference guide
    │   ├─ File organization & relationships
    │   ├─ Recommended reading order
    │   └─ Support & contact information
    ├─ BEST FOR:
    │   ├─ Understanding file structure
    │   ├─ Quick reference during work
    │   ├─ Finding specific information
    │   └─ Navigation between related files
    └─ READ TIME: 10-15 minutes
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RECOMMENDED READING ORDER (For New Users)
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STAGE 1 - GET ORIENTED (30 minutes total)
  Step 1: Read this file (FILE_MANIFEST.txt) - 10 min
  Step 2: Read SYSTEM_OVERVIEW.txt - 20 min
  → You now understand: what this is, how files relate, where to find info

STAGE 2 - QUICK START (45 minutes total)
  Step 3: Read QUICKSTART.md (Installation + Running) - 15 min
  Step 4: Run multi_drone_radar_main - 15 min
  Step 5: Review generated plots using QUICKSTART.md guide - 15 min
  → You now have: working simulation, understand basic outputs

STAGE 3 - UNDERSTAND RESULTS (60 minutes total)
  Step 6: Read README.md (sections 1-6) - 30 min
  Step 7: Study output plots in detail - 15 min
  Step 8: Experiment with parameter changes - 15 min
  → You now understand: system architecture, radar theory, plot meanings

STAGE 4 - ADVANCED STUDY (90+ minutes total)
  Step 9: Run beamforming_analysis.m - 5 min
  Step 10: Read BEAMFORMING_REFERENCE.txt - 45 min
  Step 11: Read README.md (sections 7-9) - 20 min
  Step 12: Modify code & experiment with advanced features - 20 min
  → You now understand: beamforming theory, adaptive algorithms, research
directions

Total estimated time: 4-5 hours (stages 1-3), additional 2-3 hours for stage 4
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FILE DEPENDENCY GRAPH
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File Dependencies:

  multi_drone_radar_main.m
     ├─ REQUIRES: multiDroneMotion.m (helper function)
     ├─ REQUIRES: Phased Array System Toolbox
     ├─ REQUIRES: Signal Processing Toolbox (recommended)
     ├─ PRODUCES: 7 figures (stored in MATLAB workspace)
     └─ DOCUMENTS: README.md, QUICKSTART.md, BEAMFORMING_REFERENCE.txt
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  multiDroneMotion.m
    ├─ REQUIRES: MATLAB core (no toolbox)
    ├─ CALLED BY: multi_drone_radar_main.m (4 times per pulse)
    ├─ NO EXTERNAL FILES NEEDED
    └─ DOCUMENTS: README.md (section 3.2), BEAMFORMING_REFERENCE.txt

  beamforming_analysis.m
    ├─ REQUIRES: Phased Array System Toolbox
    ├─ INDEPENDENT (doesn't call other scripts)
    ├─ PRODUCES: 9 figures (stored in MATLAB workspace)
    └─ DOCUMENTS: README.md, QUICKSTART.md, BEAMFORMING_REFERENCE.txt

  README.md
    ├─ DOCUMENTS: All .m files, system architecture, physics theory
    ├─ REFERENCES: BEAMFORMING_REFERENCE.txt, QUICKSTART.md
    └─ BEST VIEWED: After running simulation (understand output)

  QUICKSTART.md
    ├─ DOCUMENTS: How to run, parameter tuning, troubleshooting
    ├─ REFERENCES: FILE_MANIFEST.txt, README.md
    └─ BEST VIEWED: Before first run (for installation)

  BEAMFORMING_REFERENCE.txt
    ├─ DOCUMENTS: Detailed beamforming mathematics & theory
    ├─ REFERENCES: README.md section 8, research papers
    └─ BEST VIEWED: During detailed study phase

  SYSTEM_OVERVIEW.txt
    ├─ DOCUMENTS: Project summary, file descriptions, specifications
    ├─ REFERENCES: All other files (cross-links)
    └─ BEST VIEWED: First (orientation document)

  FILE_MANIFEST.txt (this file)
    ├─ DOCUMENTS: File structure, relationships, recommended reading
    ├─ REFERENCES: All other files
    └─ BEST VIEWED: Early (navigation guide)

================================================================================
QUICK REFERENCE LOOKUP TABLE
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LOOKING FOR...                         GO TO FILE...
─────────────────────────────────────────────────────
System overview                        SYSTEM_OVERVIEW.txt
Installation instructions              QUICKSTART.md
How to run simulation                  QUICKSTART.md
Parameter configuration                QUICKSTART.md
Plot interpretation                    QUICKSTART.md, README.md
Beamforming theory                     BEAMFORMING_REFERENCE.txt
Radar physics & equations              README.md, BEAMFORMING_REFERENCE.txt
Simulation architecture                README.md
Troubleshooting errors                 QUICKSTART.md
File descriptions                      FILE_MANIFEST.txt
File dependencies                      FILE_MANIFEST.txt
Steering vector math                   BEAMFORMING_REFERENCE.txt
Beam pattern explanation               BEAMFORMING_REFERENCE.txt
Array gain calculation                 BEAMFORMING_REFERENCE.txt
Micro-Doppler physics                  BEAMFORMING_REFERENCE.txt, README.md
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Matched filter details                 BEAMFORMING_REFERENCE.txt
RCS model explanation                  README.md
Free-space propagation                 README.md
Advanced topics                        README.md section 8
Research directions                    README.md section 8
References & citations                 README.md section 9
MATLAB examples                        README.md
Next steps after simulation            SYSTEM_OVERVIEW.txt
Performance metrics                    SYSTEM_OVERVIEW.txt
Drone specifications                   SYSTEM_OVERVIEW.txt
Radar specifications                   SYSTEM_OVERVIEW.txt

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FILE STATISTICS
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MATLAB CODE:
  multi_drone_radar_main.m        ~1,100 lines
  multiDroneMotion.m              ~130 lines
  beamforming_analysis.m          ~700 lines
  ───────────────────────────────────────
  TOTAL CODE                      ~1,930 lines

DOCUMENTATION:
  README.md                       ~600 lines
  QUICKSTART.md                   ~450 lines
  BEAMFORMING_REFERENCE.txt       ~500 lines
  SYSTEM_OVERVIEW.txt             ~550 lines
  FILE_MANIFEST.txt               ~350 lines (this file)
  ───────────────────────────────────────
  TOTAL DOCUMENTATION             ~2,450 lines

COMBINED TOTAL                    ~4,380 lines of content

DISK USAGE:
  MATLAB code:                    ~90 KB
  Documentation:                  ~280 KB
  TOTAL:                          ~370 KB

DOWNLOAD SIZE:                    ~400 KB (all files together)

================================================================================
SYSTEM REQUIREMENTS
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MINIMUM REQUIREMENTS:
  OS                  : Windows 10+, macOS 10.13+, Linux (Ubuntu 18.04+)
  MATLAB Version      : R2023a or later
  Memory (RAM)        : 8 GB (4 GB minimum for quick test)
  Storage             : 2-5 GB (for full 10-second simulation)
  Processor           : Any modern multi-core CPU (i5+, Ryzen 5+, etc.)

REQUIRED TOOLBOXES:
  - Phased Array System Toolbox (core requirement)
  - Signal Processing Toolbox (recommended)

OPTIONAL TOOLBOXES:
  - Parallel Computing Toolbox (for GPU acceleration - not implemented)
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- Statistics and Machine Learning Toolbox (for advanced analysis)

RECOMMENDED SYSTEM:
- 16 GB RAM (for comfortable multi-tasking)
- SSD storage (faster file access)
- Dedicated GPU (optional, for future acceleration)

PERFORMANCE BY CONFIGURATION:

```
  Quick Test (Config A):    2-3 min runtime    (Ndrones=2, duration=1s)
  Balanced (Config B):      5-7 min runtime    (Ndrones=4, duration=3s)
[DEFAULT]
  Full 10s (Config C):      10-15 min runtime  (Ndrones=4, duration=10s)
  High-Res (Config D):      20-30 min runtime  (Ndrones=4, 16×16 array)
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SUPPORT & TROUBLESHOOTING
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IF YOU GET AN ERROR:
1. Check console output for specific error message
2. Review QUICKSTART.md "Troubleshooting Guide" section
3. Verify Phased Array Toolbox is installed: ver
4. Try Quick Test configuration first (Config A)
5. Check file paths and working directory

IF YOU NEED HELP:
1. Read inline comments in .m files (marked with %%)
2. Review relevant documentation sections (see lookup table above)
3. Search for error message in README.md or QUICKSTART.md
4. Enable MATLAB debugging: Add fprintf() statements in code
5. Try reducing problem size (fewer drones, shorter duration)

IF YOU WANT TO EXTEND:
1. Start with multi_drone_radar_main.m as template
2. Understand simulation loop structure (main for-loop)
3. Modify one section at a time, test after each change
4. Use beamforming_analysis.m for visualization of changes
5. Document your modifications in comments

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COPYRIGHT & LICENSE
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This simulation system is provided for:
- ✓ Educational purposes
- ✓ Research and development
- ✓ Academic study
- ✓ Non-commercial use

Status: Ready for use
Generated: 2026
Maintained by: Educational/Research Community

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END OF FILE MANIFEST
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Questions? Start with SYSTEM_OVERVIEW.txt or QUICKSTART.md
Ready to run? Execute: >> multi_drone_radar_main

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